



LATHER HIGH SCHOOL, KARNAL

Website Redesign & Content Management System Portal Project Final Handover Documentation

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Target Client:	Board of Directors, Lather High School, Karnal
Prepared By:	Web Development & Solutions Team
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1. Executive Project Summary

This project deliverables document details the final handoff specifications for the premium, custom-designed institutional portal and CMS administrative center built for **Lather High School, Karnal**. The redesign aimed to align the school's digital footprint with its premium brand reputation, featuring dynamic information hubs, interactive directories, and a secure payment desk.

Key goals achieved in this release include a highly engaging, video-enabled home page layout, an optimized search and bio directory for faculty and alumni, instant fee structure updates and digital payments, and a bespoke admin center allowing administrative staff to securely update counters, publish events, upload newsletters, and edit page elements dynamically.

2. Technical Stack & System Architecture

The architecture is divided into a decoupled client-server structure, prioritizing security, lightning-fast PageSpeed index, and operational autonomy.

Component	Technology Profile	Role & Advantage
Frontend framework	Next.js 15 (TypeScript, App Router)	Delivers advanced SEO performance, quick server rendering, and page-caching configurations.
Styling engine	Tailwind CSS v4	Highly utility-first, modern responsive style frameworks with custom brand theme tokens.
Animation engine	Framer Motion & GSAP	Powers premium scroll transitions, modal popups, and slider components.
Backend runtime	Node.js with Express & TS	Decoupled REST API router implementing JWT token validation and secure file uploading.
Database & ORM	Prisma Client, SQLite	Flexible ORM. Local development uses SQLite; instantly convertible to PostgreSQL in production via environment params.

3. Directory Structure & Key Public Routes

The public site implements a clean, logical routing layout optimized for different user personas (prospective parents, alumni, current students, faculty).

- **Homepage (/)** : Houses the high-end overlay video hero, achievements stats counters (Years of Legacy, Enrolled Students, Expert Faculty, National Awards), Principal's welcome message, upcoming events calendar, news card rows, and testimonials slider.
- **About Page (/about)**: Introduces the school's history, mission, vision, and core educational philosophy, including interactive timelines of key milestones and an infrastructure slider.
- **Faculty Directory (/faculty)**: Includes active search indexers and department selectors. Prompts bios within modular cards directly populated from the backend API.
- **Alumni Directory (/alumni)**: Showcases batch selectors and career achievement logs. Filters alumni card profiles with linked profiles.
- **Blog Portal (/blog)**: Retrieves articles dynamically. Incorporates secondary categories, featured banners, publication dates, and HTML markup rendering.
- **Secure Payment Desk (/payment)**: Provides downloadable UPI QR codes, fee structure PDFs, offline bank transfer details, and payment FAQ accordions.

4. Administrative CMS Portal & Editor Controls

Administrative staff can access the CMS control panel securely via **/admin**. Once logged in, users are presented with an active sidebar navigation connecting separate editor interfaces:

- **Sidebar Dashboard Editor**: Mutate home counters in real-time, modify welcome headings, adjust hero subtext, and save updates instantly.
- **Faculty Directory Editor**: Create, edit, or delete faculty records. Upload new photo files directly via the built-in media uploader.
- **Alumni Portal Editor**: Update career achievements, manage alumni spotlights, and verify registration requests.
- **Blog Publisher Workspace**: Draft news articles with a built-in rich text composer. Set items as draft or toggle publish status dynamically.
- **Payment Settings Panel**: Upload updated term structure PDFs and update QR code media paths without modifying code.
- **Database Backup Desk**: Allows immediate exports and imports of site settings in JSON formats for absolute data safety.

5. Database Model Reference Schema

The relational SQLite database (mapped using Prisma Client) contains the following tables and data structures:

Table Name	Fields & Types	Purpose / Constraints
Admin	id: Int (PK) username: String (Unique) password: String (bcrypt)	Stores administrative credentials for CMS login.
Faculty	id: Int (PK) name: String department: String qualification: String experience: String photoUrl: String bio: String	Tracks faculty directory items. Uploaded files map to static folders.
Alumni	id: Int (PK) name: String batch: String currentPosition: String company: String achievement: String photoUrl: String linkedin: String?	Records alumni profiles and external links.
Gallery	id: Int (PK) type: String (image/video) url: String category: String orderIndex: Int	Tracks campus life media items in categorical collections.
Blog	id: Int (PK) title: String content: String slug: String (Unique) category: String featuredImage: String draft: Boolean publishedAt: DateTime	Manages news entries and blog publications.
Event / News	id: Int (PK) title: String date: String description/content: String location/imageUrl: String	Feeds calendar timetables and media announcements.
PageContent	key: String (PK, Unique) value: String (JSON payload)	Flexible key-value storage for page sections, text blocks, and SEO metadata.

6. Backend REST API Endpoints

A decoupled REST API running on Port 5000 processes client queries. JWT verification coordinates write routes.

Method / Route	Auth	Description
POST /api/auth/login	Public	Accepts credentials, returns JWT token (24h validity).
GET /api/auth/verify	Bearer	Validates active session and returns admin context.
GET /api/cms/page/:key	Public	Fetches JSON content blob for home, about, payment, etc.
POST /api/cms/page/:key	Bearer	Saves updated editor settings inside PageContent.
POST /api/cms/upload	Bearer	Multer file upload (image/video/PDF) up to 50MB.
GET /api/cms/{faculty alumni blog event news}	Public	Retrieves list logs (ordered, filtered).
POST/PUT/DELETE /api/cms/.../:id	Bearer	CRUD handlers for modifying dynamic items.